MULTICLASS SENTIMENT ANALYSIS

OF PHONE REVIEWS

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WHAT

IS SENTIMENT ANALYSIS

- Sentiment Analysis (SA), also known as opinion mining, is a Natural Language Processing (NLP) technique used to determine the sentiment behind a piece of data—whether it is positive, negative, or neutral forming.
- It is critical in e-commerce, where customer reviews contain valuable insights into product quality, customer preferences, and potential areas for improvement.
- Businesses use SA to make data-driven decisions about product development, marketing, and customer service

amazon UNLOCKED PHONES REVIEW

Brigitte Schwartz

★★★★★

This phone is a game-changer for me. It offers me monthly dealership and at the same time I have the newest edition too! How amazing!!

Rufus Stewart

★★★★★

I'm really impressed with this unlocked phone! Setup was super easy, and it connected seamlessly with my carrier. The battery life is fantastic, lasting more than a day even with heavy use.

Korina Villanueva

★

I really wanted to like this phone, but it's been a frustrating experience. It keeps dropping calls, and the signal reception is poor, even though it's supposed to work on all networks.

Taylor Alonso

★★★★★

Overall, this is a solid unlocked phone for the price. It performs well for everyday tasks, and I appreciate the freedom of using any carrier. The display is bright, and the sound quality is good.

OBJECTIVE (PREVIOUS)

- **Classifying Customer Reviews:** Using machine learning and deep learning models to classify the sentiment of reviews as positive, neutral, or negative.
- **Model Comparison:** Comparing the performance of various models like DistilBERT, RoBERTa, Logistic Regression, and Random Forest to determine which model performs best in understanding and predicting customer sentiments.
- **Extracting Insights:** Analyzing patterns in customer feedback to provide insights into product quality, customer satisfaction, and potential areas for improvement.
- **Data-Driven Decision Making:** Empowering businesses to use sentiment analysis for improving product offerings, customer service, and marketing strategies based on customer reviews.

OBJECTIVE (UPDATED)

- Incorporate additional models for multiclass sentiment analysis:
    - ELECTRA: An efficient transformer-based model for text analysis.
    - LSTM: A deep learning model designed for sequential data processing.
    - Random Forest: A traditional ensemble-based machine learning model.
- Compare performance of all models, including:
- DistilBERT and Logistic Regression (from the mid-term phase).
- Evaluate models based on:
- Accuracy, Precision, Recall, and F1-score.
- Conduct a comprehensive analysis to determine the most effective model for sentiment classification.

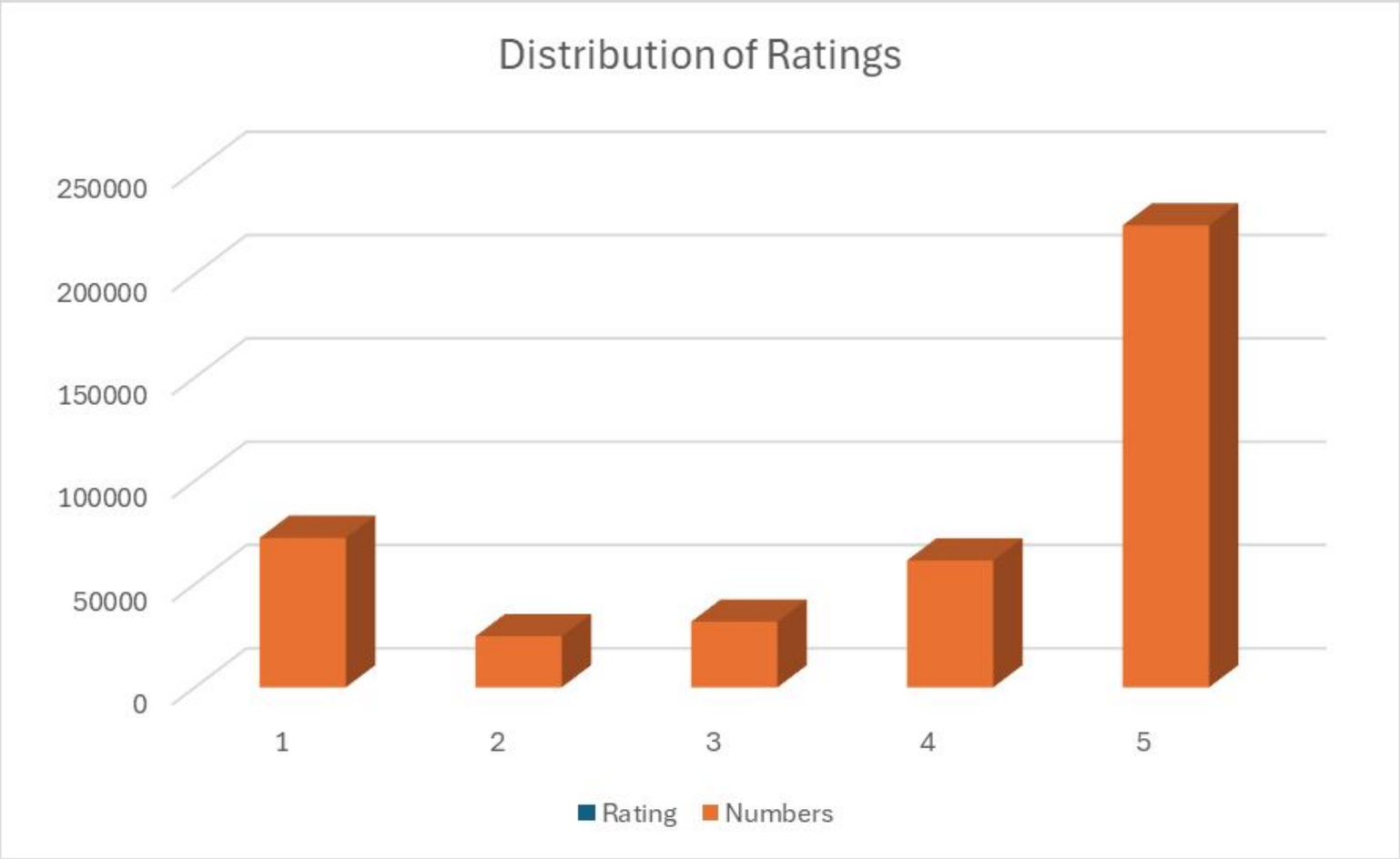
DATASET

- The Amazon Unlocked Mobile Phones Dataset is a collection of over 400,000 reviews of unlocked mobile phones sold on Amazon. The reviews were left by verified customers who purchased these phones.
- Dataset Source: Kaggle - Amazon Reviews for Unlocked Mobile Phones
- This dataset includes information such as review IDs, reviewer IDs, product IDs, ratings, review titles, review text, and timestamps of when the reviews were written.
- It also provides data on verified purchases and helpfulness votes given to the reviews, indicating their value to other users.
- This dataset is ideal for analyzing customer sentiments, product ratings, and building machine learning models for review classification, sentiment analysis, and product recommendations.

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes
0	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	5	I feel so LUCKY to have found this used (phone...	1.0
1	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	4	nice phone, nice up grade from my pantach revu...	0.0
2	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	5	Very pleased	0.0
3	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	4	It works good but it goes slow sometimes but i...	0.0
4	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	4	Great phone to replace my lost phone. The only...	0.0
...	...	...	...	...	...	...
413835	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	5	another great deal great price	0.0
413836	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	3	Ok	0.0
413837	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	5	Passes every drop test onto porcelain tile!	0.0
413838	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	3	I returned it because it did not meet my needs...	0.0
413839	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	4	Only downside is that apparently Verizon no lo...	0.0

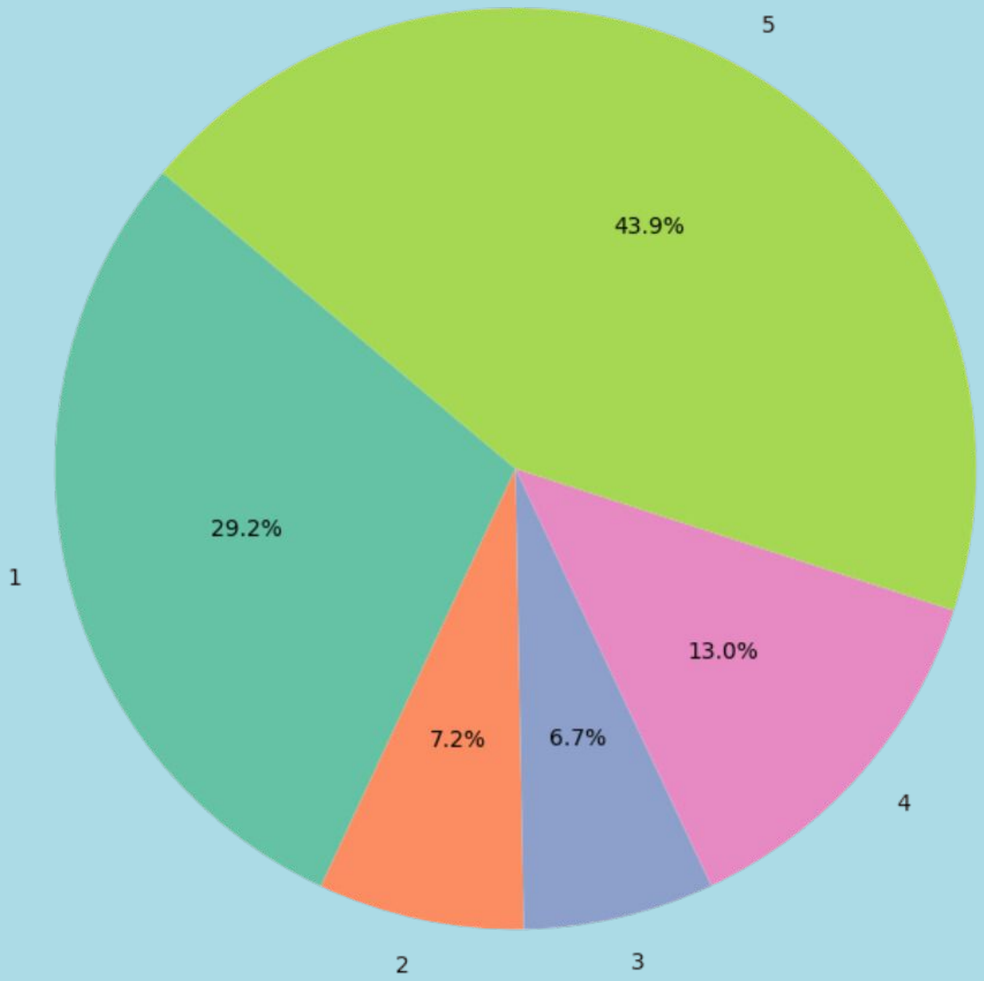
413840 rows × 6 columns

EXPLORATORY DATA ANALYSIS




```
Rating
1     72350
2     24728
3     31765
4     61392
5    223605
Name: count, dtype: int64
```

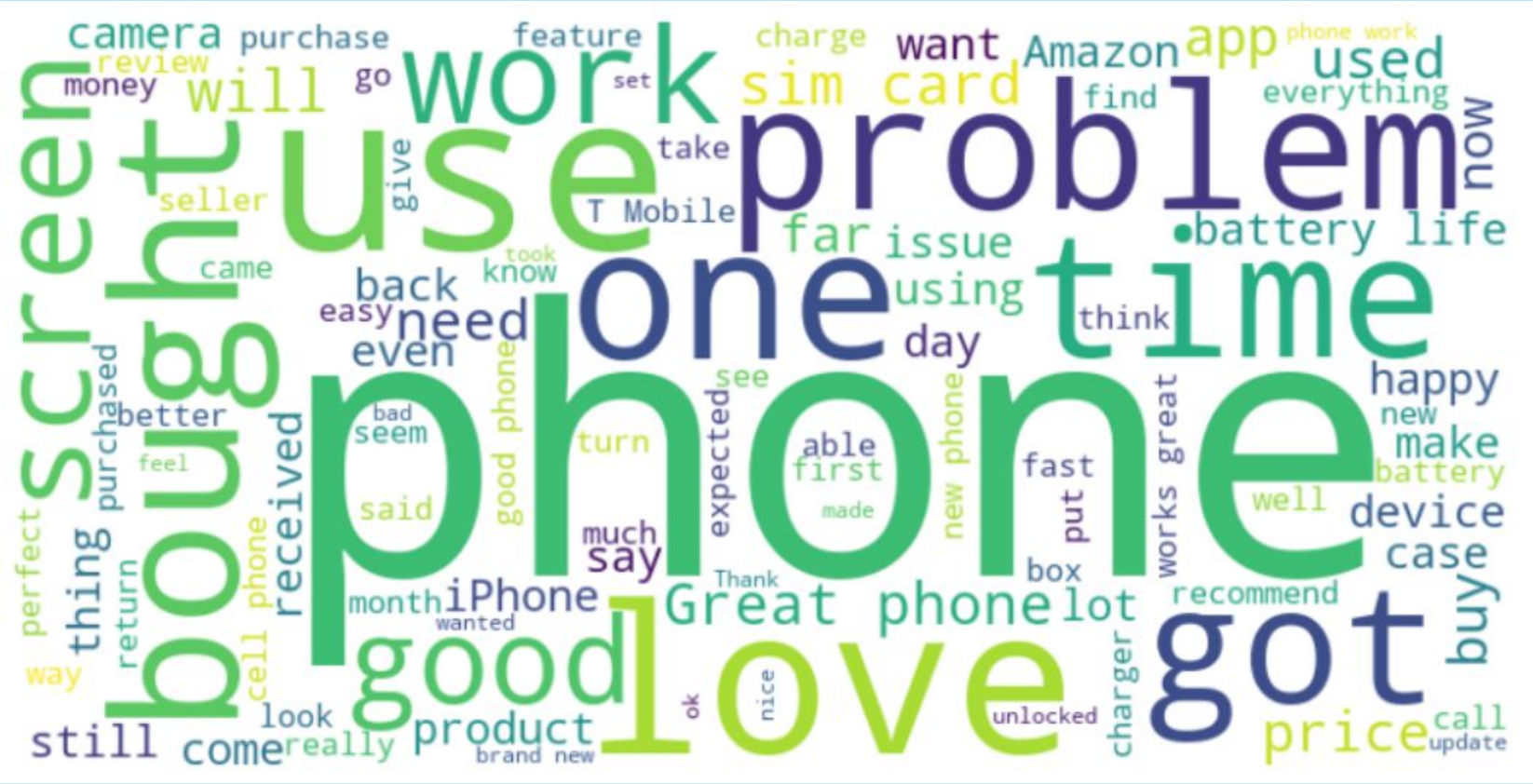
EXPLORATORY DATA ANALYSIS




```
Rating
1    29.160209
2     7.221482
3     6.701343
4    13.065949
5    43.851017
Name: Review Votes, dtype: float64
```

VISUALIZATION - WORD CLOUD

Generated a word cloud to visualize frequent words in the reviews after preprocessing (e.g., "phone", "use", "love", “work”, “problem”, “screen”, “bought”)



EXPLORATORY DATA ANALYSIS

```
df.isnull().sum()
```

	0
Product Name	0
Brand Name	65171
Price	5933
Rating	0
Reviews	70
Review Votes	12296

```
df.info()
```

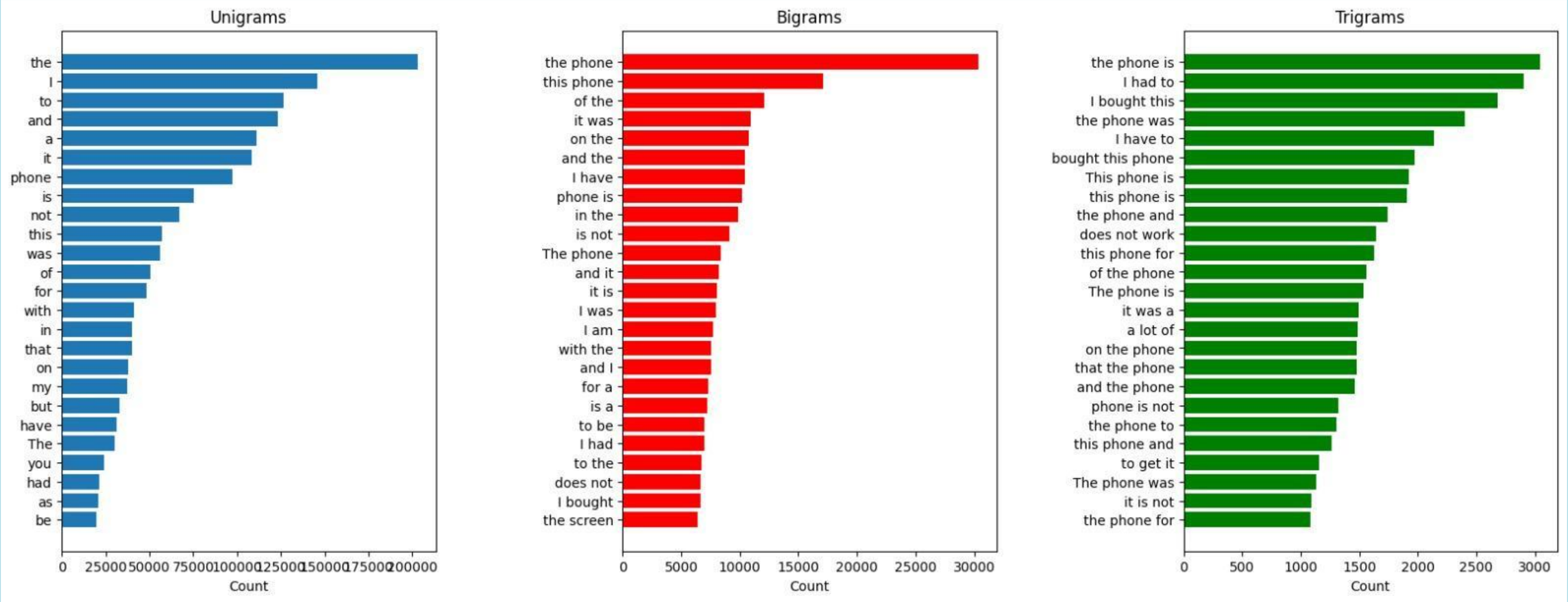
```
<class 'pandas.core.frame.DataFrame'>
Index: 413770 entries, 0 to 413839
Data columns (total 7 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   Product Name          413770 non-null  object
 1   Brand Name            413770 non-null  object
 2   Price                 413770 non-null  float64
 3   Rating                413770 non-null  int64
 4   Reviews               413770 non-null  object
 5   Review Votes          413770 non-null  float64
 6   preprocessed_reviews  413770 non-null  object
dtypes: float64(2), int64(1), object(4)
```

```
df.describe()
```

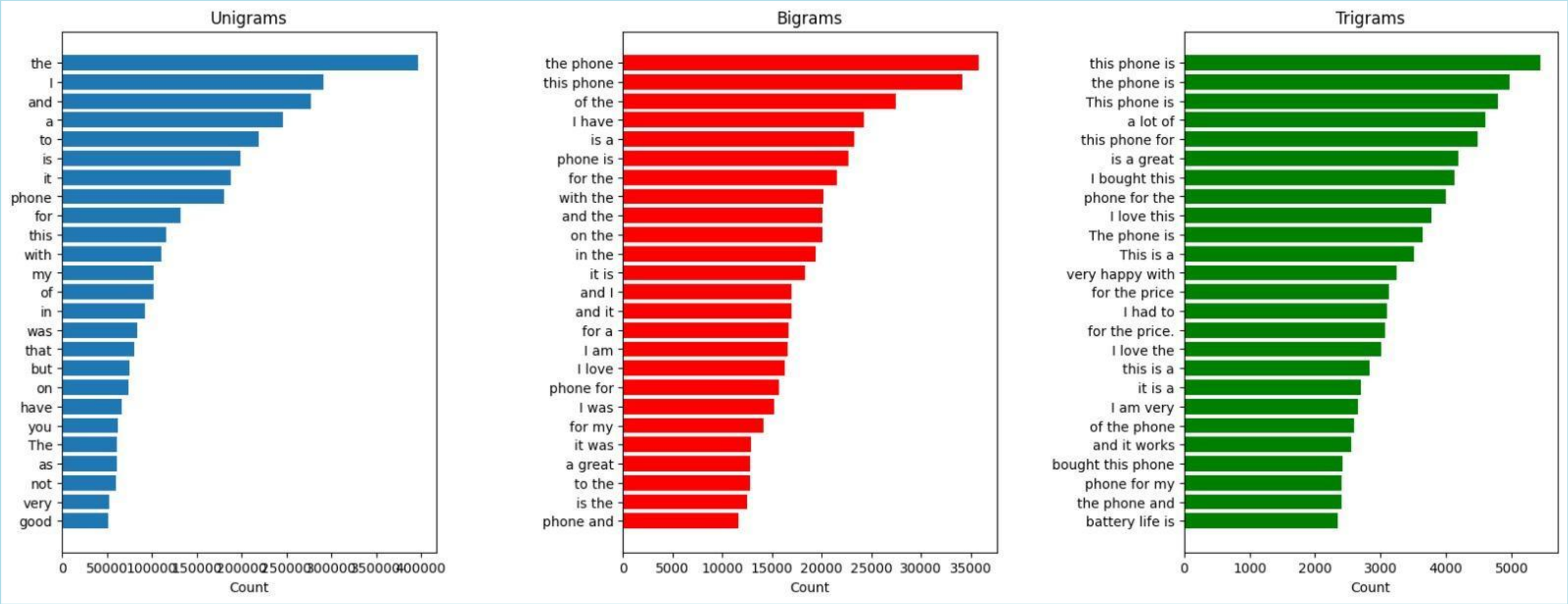
	Price	Rating	Review Votes
count	413770.000000	413770.000000	413770.000000
mean	223.616407	3.819608	1.461425
std	272.395549	1.548209	9.028987
min	0.000000	1.000000	0.000000
25%	78.000000	3.000000	0.000000
50%	139.950000	5.000000	0.000000
75%	269.100000	5.000000	1.000000
max	2598.000000	5.000000	645.000000

TEXT ANALYSIS (UNIGRAMS, BIGRAMS, TRIGRAMS)

- Unigrams: Shows the most frequent individual words in the dataset (e.g., "the," "phone," "I").
- Bigrams: Demonstrates the most frequent pairs of words (e.g., "the phone," "this phone").
- Trigrams: Displays common three-word combinations (e.g., "this phone is," "the phone is").



TEXT ANALYSIS (UNIGRAMS, BIGRAMS, TRIGRAMS)



APPLIED MODELS (LLM) :

- **DistilBERT** is a smaller, faster version of BERT (Bidirectional Encoder Representations from Transformers), designed to perform well while being more efficient. It uses transformers, a type of deep learning architecture, to understand the context of words in a sentence by analyzing them in both directions (from left to right and right to left). This makes it powerful for tasks like sentiment analysis, where understanding the nuances and context of words is critical. For the training, 16 gb text data is used which took almost 3.5 days. It has half of the parameters than BERT.
- ~~**RoBERTa (Robustly Optimized BERT Pretraining Approach)**~~ is an improvement on the original BERT model. It removes certain training limitations from BERT and trains longer on more data, resulting in a more robust and accurate model. RoBERTa is highly effective at understanding language, making it excellent for sentiment analysis tasks where subtle word differences can affect classification. For the training, 160 gb of data is used which took 4 to 5 days which is more than BERT. The parameters are same as BERT

APPLIED MODELS (LLM) :

- **ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately)** is a transformer-based model optimized for efficiency. Unlike traditional masked language models, ELECTRA uses a generator-discriminator approach during pretraining, making it faster and more accurate for downstream tasks like sentiment analysis. This efficiency enables robust text classification with fewer computational resources.

APPLIED MODELS (ML) :

- **Logistic Regression** is a traditional machine learning algorithm used for binary or multiclass classification tasks. It is relatively simple and fast, and works by estimating probabilities based on input features, then classifying the results into categories. In sentiment analysis, it would classify reviews as positive or negative based on word features extracted from the text.
- ~~**XGBoost (Extreme Gradient Boosting)**~~ is a decision-tree-based ensemble machine learning algorithm. It works by building multiple weak models (decision trees) and combining them to create a strong predictor. XGBoost is known for its speed and performance, particularly on structured data, but can also be applied to text classification tasks like sentiment analysis.

APPLIED MODELS (ML) :

- **Random Forest** is an ensemble-based machine learning model that builds multiple decision trees and combines their outputs for improved accuracy. It is particularly effective for structured data classification but has also shown strong performance in text-based sentiment analysis when features are appropriately engineered.

APPLIED MODELS (DL) :

- **LSTM (Long Short-Term Memory)** is a type of recurrent neural network (RNN) designed to process sequential data. It excels at capturing dependencies and patterns in text by retaining information over long sequences, making it highly effective for sentiment classification where the context of words is critical.

DISTILBERT

Advantages:

- Lightweight and faster than BERT while retaining performance.
- Pre-trained transformer model, ready for fine-tuning.
- Suitable for resource-constrained environments.

Disadvantages:

- May underperform in tasks requiring deep contextual understanding.
- Limited by its smaller size compared to original BERT.

ELECTRA

Advantages:

- Efficient pre-training approach (replaces token prediction with token detection).
- Outperforms BERT on several benchmarks with fewer parameters.
- Faster pre-training and inference.

Disadvantages:

- Training from scratch is resource-intensive.
- Less mature ecosystem compared to BERT and RoBERTa.

LOGISTIC REGRESSION

Advantages:

- Simple and interpretable.
- Efficient for binary classification tasks.
- Low computational cost.

Disadvantages:

- Limited to linear decision boundaries.
- Struggles with large feature sets or non-linear problems.

RANDOM FOREST

Advantages:

- Robust to overfitting with multiple decision trees.
- Handles missing data and categorical variables well.
- Suitable for both classification and regression tasks.

Disadvantages:

- Slower prediction time compared to single-tree models.
- Can be memory-intensive for large datasets.

LSTM (LONG SHORT-TERM MEMORY)

Advantages:

- Excellent for sequential data (e.g., time series, text).
- Avoids vanishing gradient issues in recurrent neural networks (RNNs).
- Can capture long-term dependencies.

Disadvantages:

- Computationally intensive.
- Requires large datasets to perform well.

METHODOLOGY

1. **Problem Formation**
2. **Raw Data Collection**
3. **Data Exploration**
   - Review vote rating
   - Distribution ratings
   - Null Value count
   - Data set statistics using info() & describe()
4. **Data Preprocessing**
   - Punctuation, special characters, stop word removal
   - Limiting letter occurances
5. **Feature Extraction**
   - Tokenization
   - Vectorization
   - Label Encoding
6. **Model Selection**
   - Distilbert
   - Logistic Regression
7. **Model Fine Tuning and Training**
8. **Evaluation**
9. **Performance analysis** ✨

DATA PREPROCESSING

1. **Library Imports**:
   - Used essential libraries like pandas, nltk, transformers, and sklearn for data manipulation, natural language processing, and model building.
2. **Data Loading:**
   - Loaded the Amazon Unlocked Mobile Phones dataset.
   - Key columns include Product Name, Brand Name, Rating, Reviews, and Review votes.
3. **Handling Missing Data**:
   - Filled missing "Brand Name" with "Missing", "Price" with 0, and "Review Votes" with 0.
   - Dropped rows with missing reviews.

DATA PREPROCESSING

4. Text Preprocessing:

- Converted text to lowercase for uniformity.
- Removed punctuation and special characters.
- Removed stop words (common insignificant words like "the", "and").
- Removed words starting with "@" (like mentions or tags).
- Handled repeated letters (e.g., “cooool” → “cool”).

DATA PREPROCESSING

Feature Size Reduction

- Compared original and preprocessed word counts.
- Achieved 18.96% feature size reduction by cleaning the text.

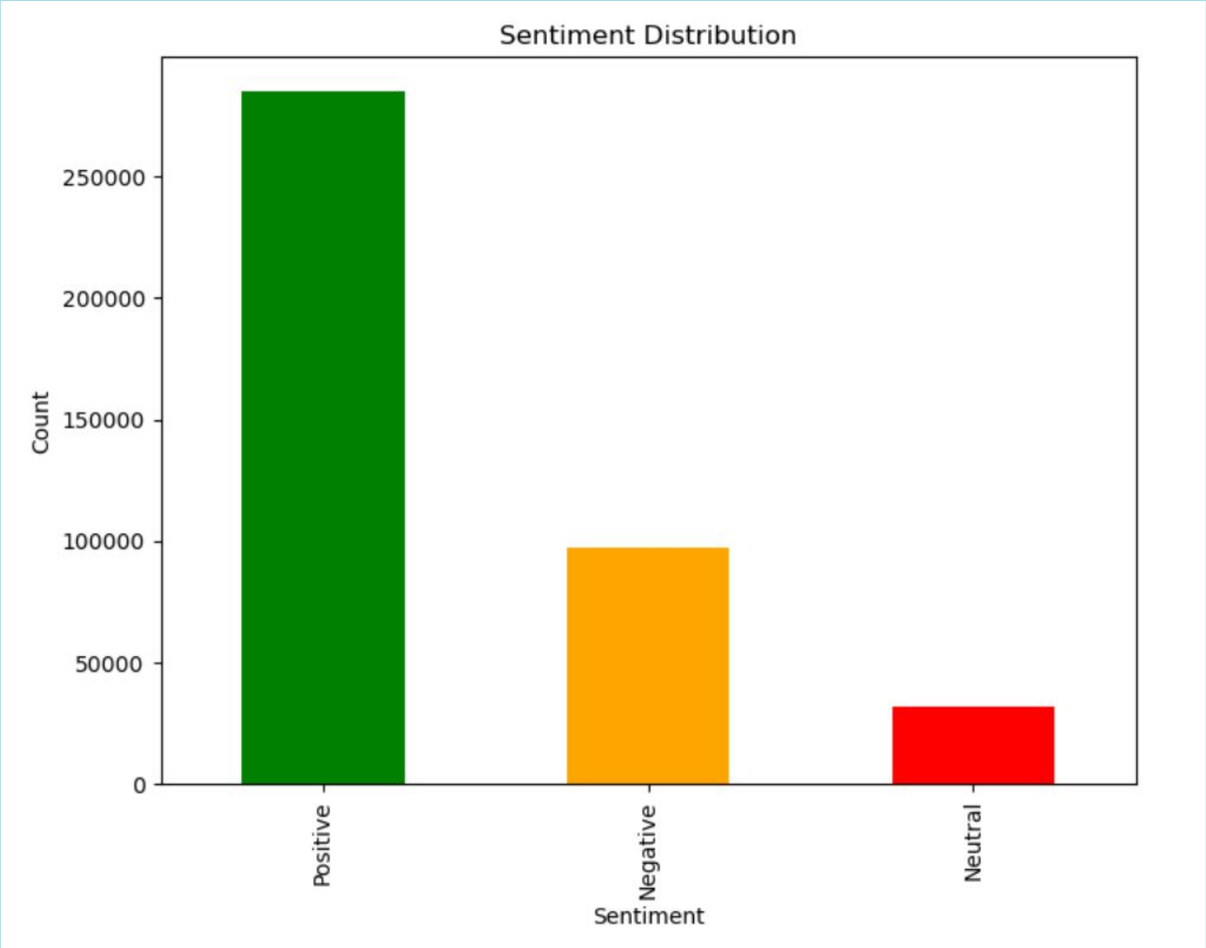
```
Original feature size: 255061
Preprocessed feature size: 206703
Percentage reduction in feature size: 18.96%
```

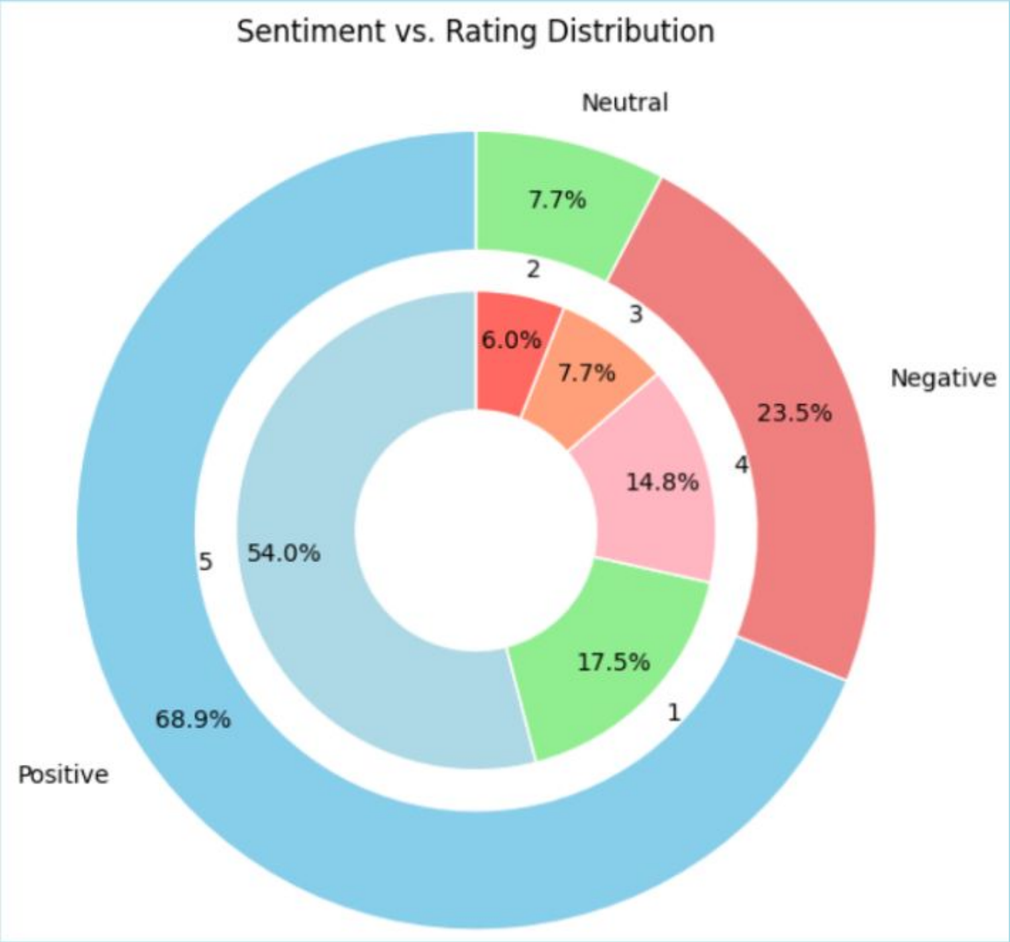
SENTIMENT LABELING

1. **Sentiment Classification: Classified reviews into**
   - Positive (Ratings 4 & 5)
   - Neutral (Rating 3)
   - Negative (Ratings 1 & 2)
2. **Sentiment Distribution**
   - Positive: 284,948 reviews
   - Negative: 97,059 reviews
   - Neutral: 31,763 reviews

	Reviews	Rating	Sentiment	Review Votes
0	Don't buy!!! It is a fake, Verified by a verif...	1	Negative	1.0
1	not worth it.. This phone just completely shut...	1	Negative	0.0
2	Didn't come with an SD Card, the Charm or the ...	3	Neutral	1.0
3	cool for the money	3	Neutral	1.0
4	Excelent phone!!!	5	Positive	0.0
5	great, sturdy case, just what I wanted.	5	Positive	0.0

SENTIMENT STATISTICS





FEATURE EXTRACTION: LLM

For multiclass sentiment classification using LLM (DistilBERT), a structured process is followed that includes tokenization, embedding generation, and fine-tuning the model.

1. **Tokenization:**
   - Used the DistilBERT tokenizer to convert text into tokens.

     tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-cased')

2. **Label Encoding:**
   - Since text labels (like "Positive," "Negative," and "Neutral") are non-numeric, they need to be converted into a numerical format that the model can understand.
   - Each unique label "Positive," "Negative," "Neutral" is mapped to a unique integer 0,1,2. It doesn't mean any order between the labels.
   - In our case, Positive='0', Negative='1', Neutral='2'

FEATURE EXTRACTION: MACHINE LEARNING

For multiclass sentiment classification using baseline machine learning model: Logistic Regression, a structured process is followed that includes vectorization, label encoding, and training the model.

1. **Text Vectorization:**

- Text data converted into numerical form using TF-IDF (Term Frequency-Inverse Document Frequency), which transforms the text into a matrix of TF-IDF features. This representation captures the importance of words in the context of the entire dataset.

2. **Label Encoding:**

- Encoded the sentiment labels for classification using LabelEncoder(), in this case, Negative='0', Neutral='1', Positive='2'.

FEATURE EXTRACTION: DEEP LEARNING

MODEL FINETUNE

1. **Create Dataset and Data Loader and set Hyperparameter:**
   - Prepare the dataset for training and validation, utilizing the tokenized inputs and encoded labels.
   - MAX_LEN = 512, TRAIN_BATCH_SIZE = 4, VALID_BATCH_SIZE = 2, EPOCHS = 2, LEARNING_RATE = 1e-05
2. **Load DistilBERT / Electra Model:**
   - Load the DistilBERT/ Electra model. Split data into train-test by 80:20 ratio and shuffled transformed to data loaders.
3. **Fine-tuning:**
   - Fine-tuning involves taking a pre-trained model that has already learned general patterns from a large corpus and adjusting it on a specific, smaller dataset for a particular task.
   - Fine-tune the model using ‘input_ids, attention_mask and target’ of the training set.
   - Used loss function “CrossEntropyLoss()” and “Adam” optimizer.

MODEL TRAINING

1. **Train-Test Split**:
   - The dataset splitted into training and testing sets. Here, 20% of the data is reserved for testing.
2. **Training**:
   - While training, the class_weight='balanced' parameter is used to help address class imbalance by automatically adjusting weights inversely proportional to class frequencies in the input data.

MODEL TRAINING: DEEP LEARNING

MODEL EVALUATION

```
DistillBERTClass(
  (l1): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): MultiHeadSelfAttention(
            (dropout): Dropout(p=0.1, inplace=False)
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072, bias=True)
            (lin2): Linear(in_features=3072, out_features=768, bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768, bias=True)
  (dropout): Dropout(p=0.3, inplace=False)
  (classifier): Linear(in_features=768, out_features=3, bias=True)
)
```

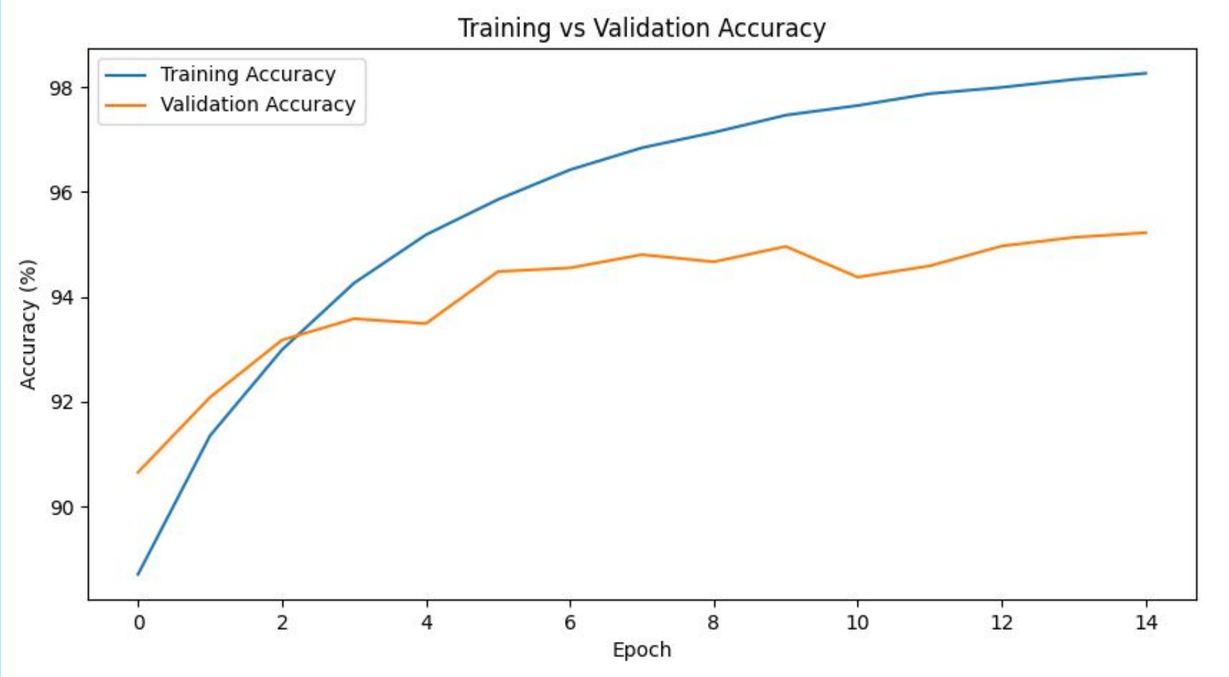
The fine-tuned /trained model is used to make predictions on the tokenized/vectorized test or validation set.

PRELIMINARY RESULTS

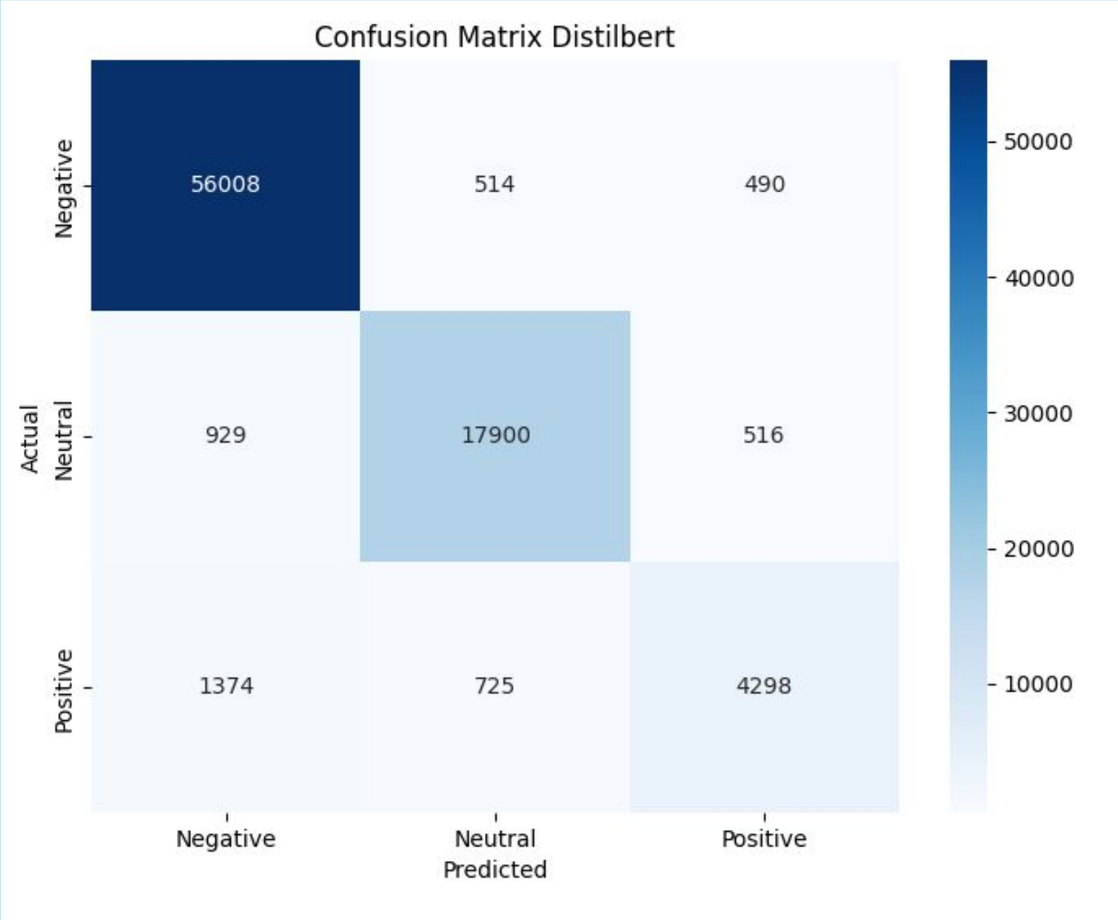
RESULTS

Model	Accuracy	Precision	Recall	F1-Score	Training Time (sec)	Testing Time (sec)
Logistic Regression	0.83	0.8912	0.8299	0.8521	31.80	0.04
Random Forest	0.94	0.9435	0.9446	0.9427	7203.18	7.11
LSTM	0.91	0.9058	0.9176	0.9116	1173.89	19.29
DistilBert	0.94	0.9567	0.9429	0.9497	72008.80	518.47
Electra	0.95	0.9601	0.9547	0.9573	34790.23	317.51

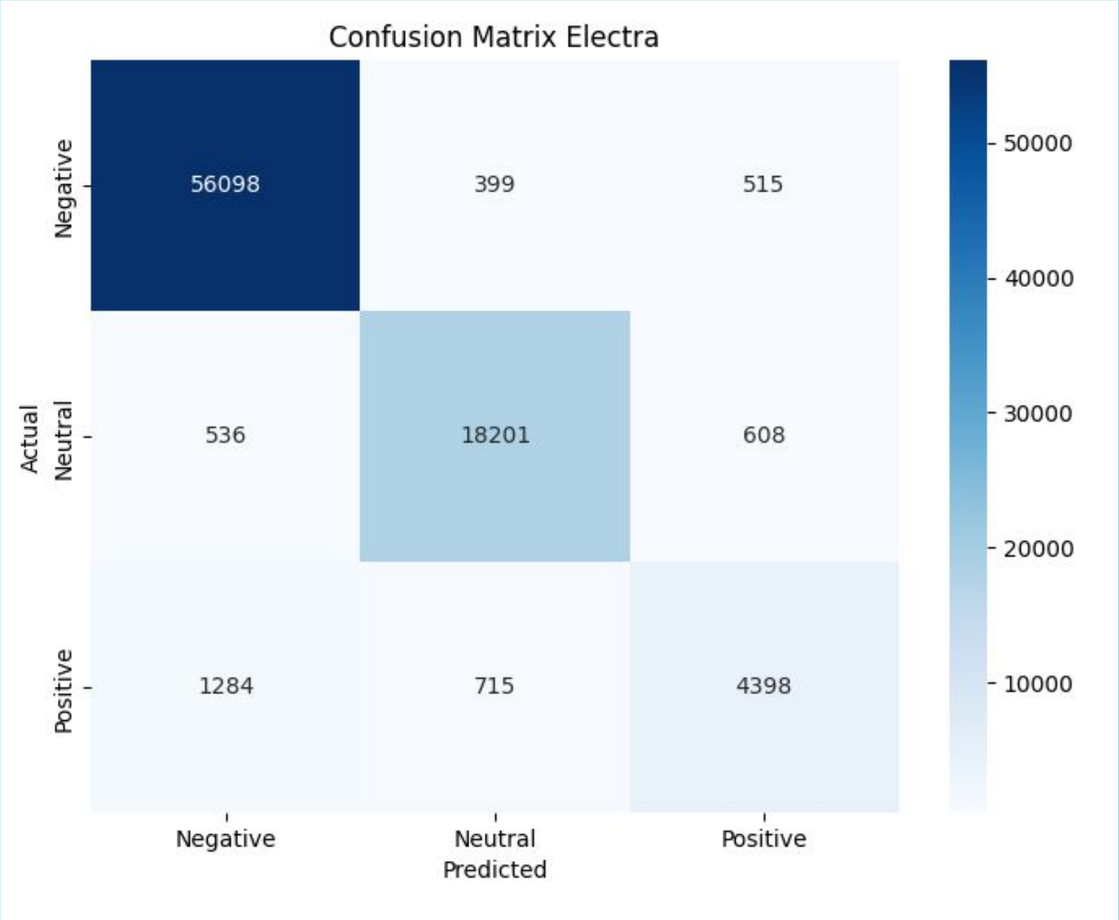
TRAINING LOSS AND ACCURACY



CONFUSION MATRIX (LLMs)

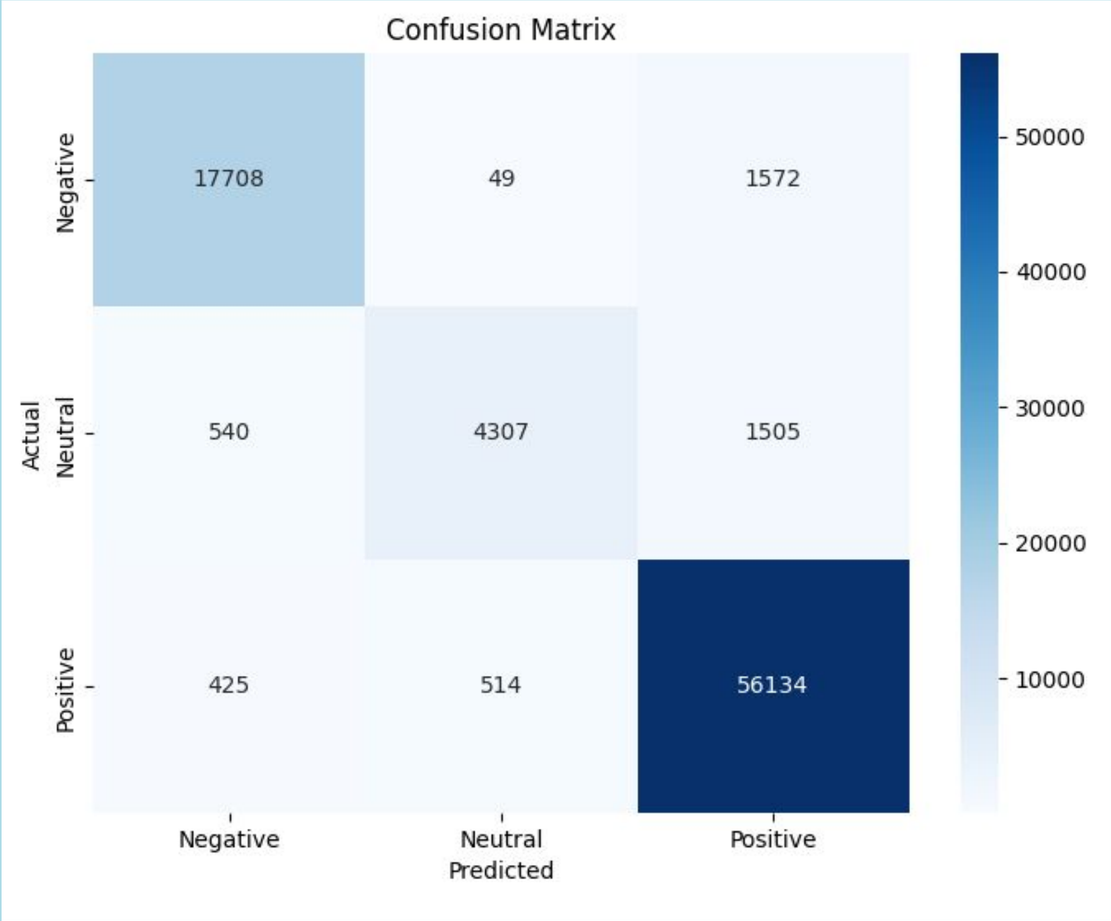


Confusion Matrix: DistilBert

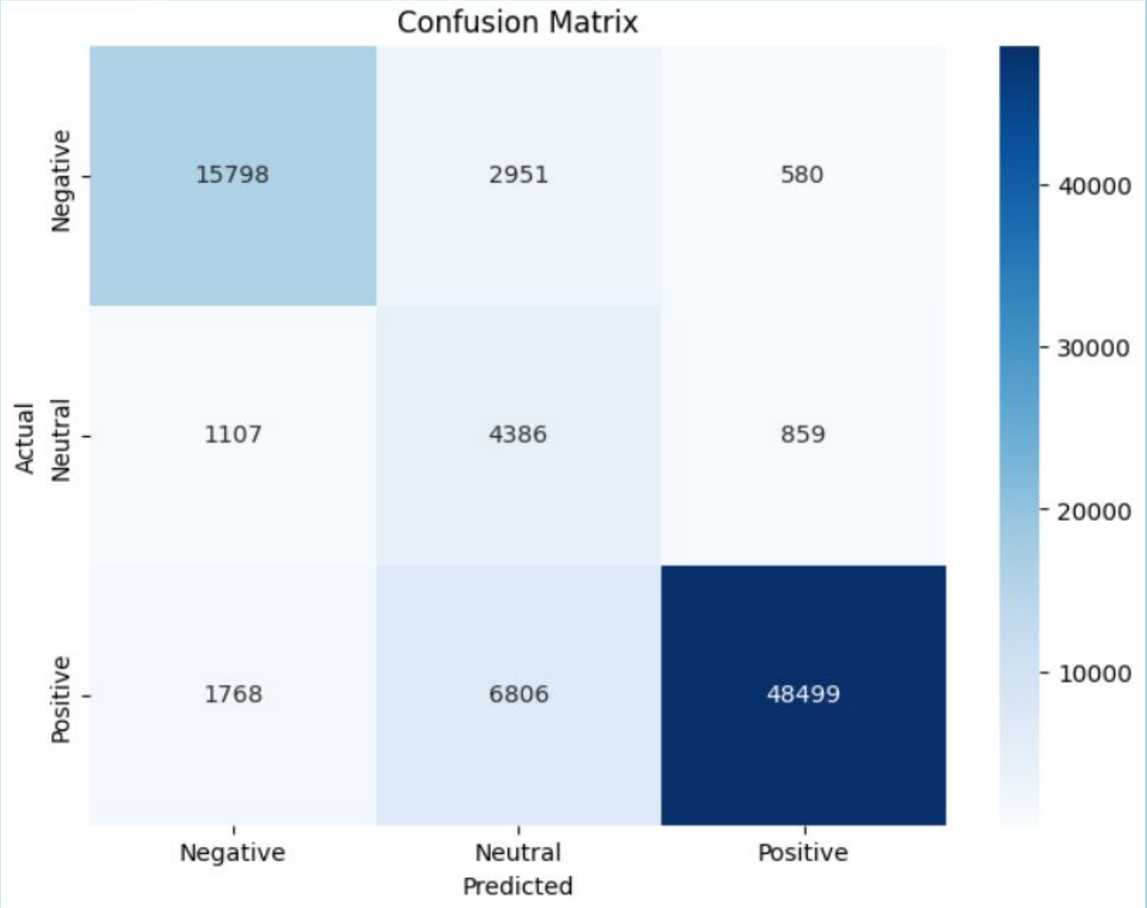


Confusion Matrix: Electra

CONFUSION MATRIX (MLs)

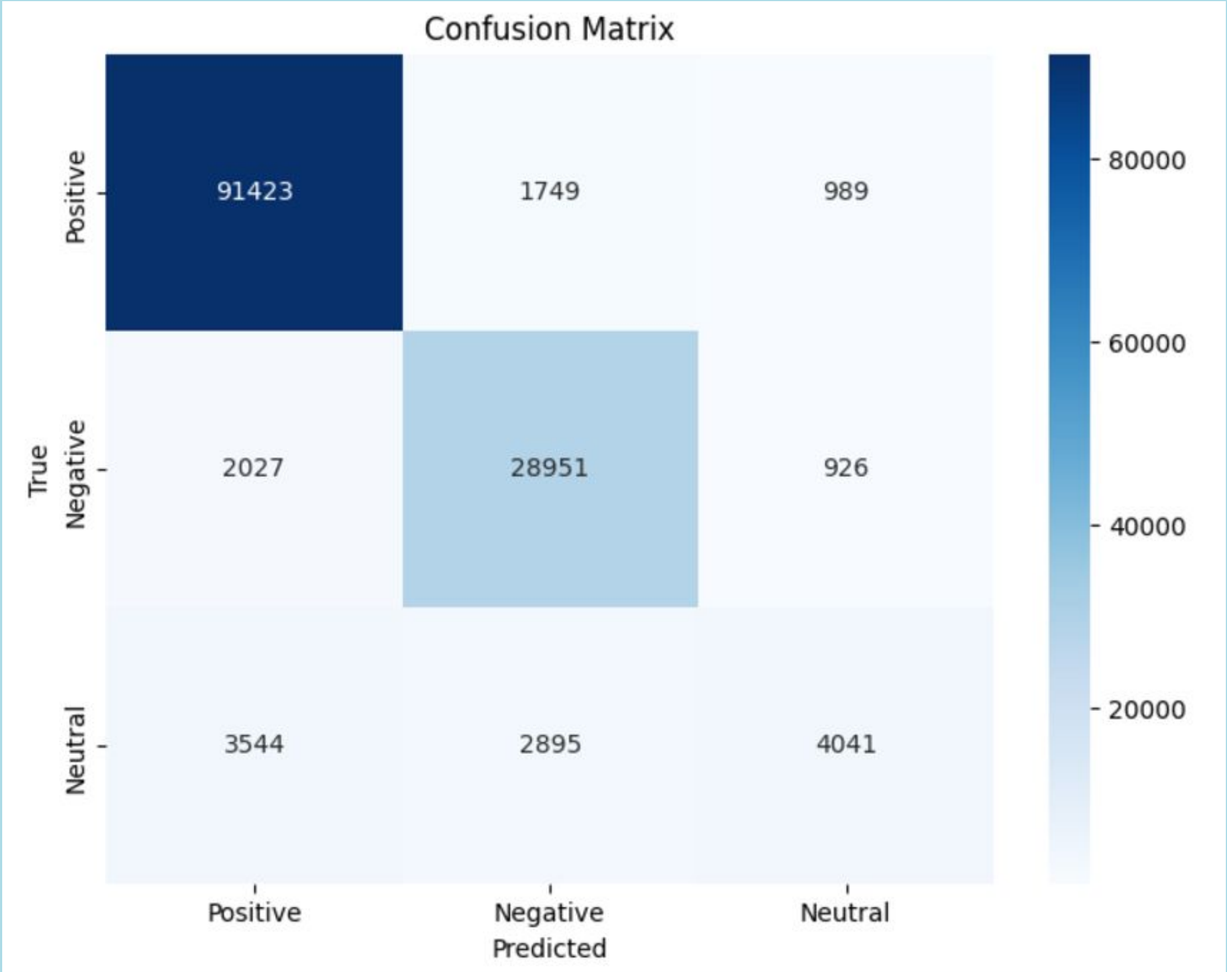


Confusion Matrix: Random Forest



Confusion Matrix: Logistic Regression

CONFUSION MATRIX (DL)



Confusion Matrix: LSTM

COMPARATIVE ANALYSIS

- Model Types:
  - Machine Learning: Logistic Regression and Random Forest are efficient but vary in performance.
  - Deep Learning: LSTM strikes a balance between performance and computational cost.
  - LLMs: DistilBERT and ELECTRA dominate in accuracy and precision, though require substantial computational resources.
- Performance:
  - ELECTRA has the highest F1-score (0.9573), followed by DistilBERT (0.9497) and Random Forest (0.9427).
  - Logistic Regression shows modest performance, suitable for baseline comparison.
- Efficiency:
  - Training Time: Logistic Regression is the fastest, while LLMs (DistilBERT, ELECTRA) are computationally expensive.
  - Testing Time: Logistic Regression and Random Forest are significantly quicker than LLMs.

NEW INSIGHTS

- **Close Performance of Advanced Models:**
  - DistilBERT and ELECTRA exhibit nearly identical high accuracy and F1-Score, indicating diminishing returns in performance improvement for the computational cost of ELECTRA over DistilBERT.
- **Trade-off in Simplicity vs. Sophistication:**
  - Logistic Regression demonstrates how simplicity sacrifices top-tier performance, but it remains viable for quick, less resource-intensive predictions.
- **Deep Learning's advantages :**
  - LSTM strikes a middle ground with good accuracy and F1-Score compared to simpler ML models and computationally heavier LLMs, suggesting it can be optimal for scenarios balancing resource constraints with performance needs.
- **Consistency Across Metrics:**
  - Models like Random Forest and LSTM exhibit balanced improvements in both Accuracy and F1-Score, showing that their ability to generalize is aligned with their ability to handle imbalanced or complex datasets.

CONCLUSION

ELECTRA offers superior performance but at a high computational cost, making Random Forest or LSTM more feasible for resource-constrained environments.

LIMITATIONS

- Limited GPU memory
- LLMs are usually larger than the ML models so it takes more computational resources.
- Finetuning time is higher for these models.
- For now we used Google Colab Pro but this is not quite efficient.
- We might need to move to some HPC resources from BSU for further exploration.

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THANK YOU!